

Development and boundaries in the field of supramolecular synthons†

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The theme of supramolecular synthons runs through all disciplines of crystal engineering. The concept of supramolecular synthons has received ever increasing attention in various fields since it was defined and introduced in 1997. This trend is manifest in the exploding number of articles published, and not only in the specialised journals, *Crystal Growth and Design* and *CrystEngComm*. Attention is being given both to supramolecular synthons, which involve strong intermolecular interactions, and to weak intermolecular interactions, for instance $C\cdots X\cdots CN$, $C-H\cdots N$, $C-H\cdots F$ or $X\cdots X$, in the interpretation and understanding of molecular aggregation in the solid state. This highlight article will give a picture of the rapid development of interest in supramolecular synthons in the last ten years together with a discussion about the boundaries of this concept.

Chemistry beyond the Molecule

“Chemistry beyond the Molecule” was the title of an article, published in 2001, in which Desiraju described the growing importance of supramolecular chemistry.¹ As Lehn has argued, intermolecular interactions play an important role in the formation of supramolecules, a role comparable to that of covalent bonds in single molecules.² Desiraju predicted the future of crystal engineering would belong to intermolecular interactions. The survey depicted in Fig. 1 shows a steady increase in the number of articles

in the field of crystal engineering during the last fifteen years. Simultaneously, the number of articles specifically dealing with supramolecular synthons, a theme running through all areas of crystal engineering, has also been steadily growing.

A supramolecular synthon is a structural unit of a supramolecule which can be formed and/or assembled by known or conceivable synthetic operations involving intermolecular interactions.³ The concept of supramolecular synthons involves the defined structural kernel of a crystal structure, which encapsulates the essence of the crystal in terms of molecular recognition. Supramolecular synthons, which involve strong intermolecular interactions, namely hydrogen bonds, had early been recognized as important intermolecular linkages. How-

ever, it has now become almost routine to discuss and analyse supramolecular synthons in terms of weak $C-H\cdots O$, $C-H\cdots N$, $C-H\cdots F$, $C-H\cdots Cl$, $CH\cdots\pi$ and $X\cdots X$ intermolecular interactions.⁴

How well do we currently understand the aggregation of molecules in solid state? Most of the intermolecular interactions are not very stable and their formation behaviour is related to and affected by small changes in the molecular structure and the crystallisation conditions. Therefore the goal should be to recognise and design synthons that are robust enough to be exchanged from one network structure to another.³ This highlight article will give a picture of the rapid development of interest in supramolecular synthons together with a discussion about the boundaries of this concept.

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† Dedicated to Prof. Roland Boese on the occasion of his 65th birthday.



Klaus Merz

Klaus Merz studied at the Universities of Cologne and Stuttgart and obtained his PhD from the University of Stuttgart under the supervision of Gerd Becker. Following postdoctoral studies in Heidelberg at Matthias Driess, he moved to the Ruhr University of Bochum. His current research is focused on the role of weak intermolecular interactions in molecular aggregation processes in solid state and precise control of crystal growth with organic molecules.



Vera Vasylyeva

Vera Vasylyeva was born in the Ukraine and moved to Germany in 2002 at the age of 17. She studied at the Ruhr University of Bochum, where she received her M Sc degree. Since 2009 she is working on her PhD theses under the supervision of Klaus Merz on the topic of weak intermolecular interactions combining syntheses with crystallographic analysis of low melting aromatic compounds.

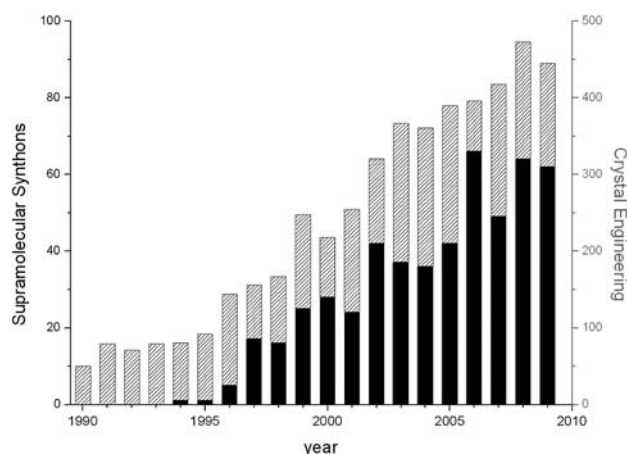


Fig. 1 A survey of articles on the topic of crystal engineering and supramolecular synthons from 1990 to 2010.

Applications

Depending on their properties, supramolecular synthons have evolved to become a useful tool for synthesising materials derived from molecular building blocks. Intermolecular interactions which are strong enough to allow the formation of structurally robust units of molecules with specific geometries are important. Weak individual interactions are also useful for synthetic flexibility and the functional diversity of desired materials.⁵ One approach to managing hydrogen bonds between supramolecular synthons effectively is to generate arrays of multiple hydrogen bonds. For example, triple hydrogen-bonding motifs^{6–8} (DAD, Fig. 2) and quadruple hydrogen-bonding motifs⁹ in combination with fullerene allow the desired construction of functionalised fullerene-based architecture. Conjugated C₆₀ units are used as building blocks for optoelectronic devices, namely light-emitting diodes (LEDs)¹⁰ or perylene bisimides (PERYs). Furthermore, tetraalkoxy-PERY-systems, with their three-point hydrogen-bonding assemblies,¹¹ and diaminopyridine C₆₀ are very useful for the generation of steady, rapid, and highly stable anodic photocurrents.

The concept of supramolecular synthons allows us to modulate analogues of hitherto unknown networks. Using weak C–H...N hydrogen bonding interactions, the two-dimensional hexagonal network structure of 2,4,6-triethynyl-1,3,5-triazine reveals the formation of a graphyne-like network with parallel stacking of the layers. The layers in the crystal packing

are determined by short C(sp)–H...N contacts with C–H...N angles of about 180°. All of the nitrogen and hydrogen atoms participate in the synthon forma-

tion, to generate a robust two-dimensional array (Fig. 3).¹² In contrast to the crystal packing of molecules, the self-assembly processes on the surface allow the addressing of individual units. Carboxylic acid groups such as trimesic acid and 1,3,5-benzenetribenzoic acid are well known synthons for a promising route to functional nanostructures with molecular self-assembly on surfaces.^{13–17} An access to the fine-tuning of surfaces of thin film can be achieved simply by adjusting intermolecular forces. Preferential binding between carboxylic acids and tetrapyrroldiporphyrin moieties can be exploited for the construction of self-assembled monolayers with stable hydrogen bonds on gold surface.¹⁸

Yokoyama used cyanophenyl substituents to control the molecular aggregation of porphyrins on gold surface. The characteristic intermolecular interactions between the cyano groups are exploited to

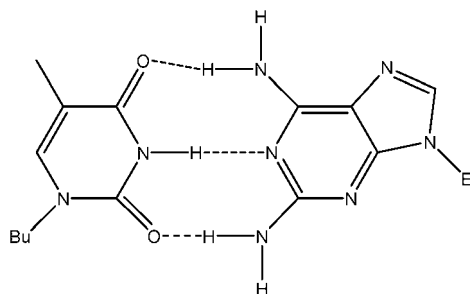


Fig. 2 Hydrogen-bonding motif in which amide (proton donor site, D) and pyridine (proton acceptor site, A) moieties are arrayed in the DAD sequence.

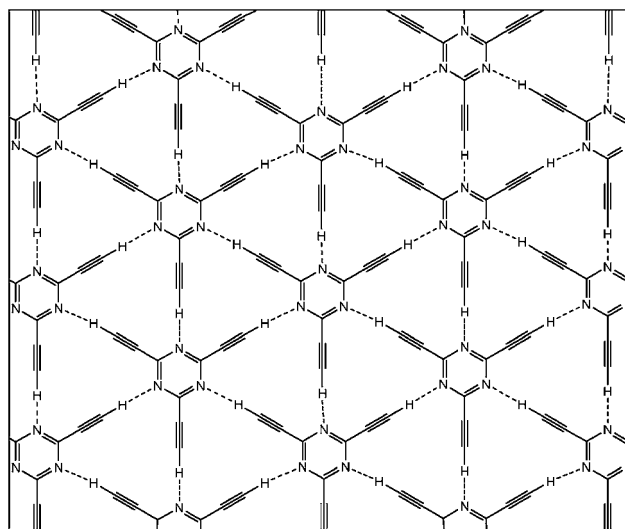


Fig. 3 2,4,6-Triethynyl-1,3,5-triazine forms a graphyne-like network with C(sp)–H...N contacts and C–H...N angles of about 180°.

influence the aggregation of the absorbed molecules forming trimers or extended wire-like structures. High-resolution STM images show the selective aggregation results from the antiparallel or cyclic configuration of the cyano substituents in relation to each other.¹⁹

Catemer and dimer $\text{HOC}=\text{O}\cdots$ $\text{HOC}=\text{O}$ and $\text{C}=\text{O}\cdots\text{H}-\text{N}$ motifs

The flexibility of supramolecular synthons depends on small differences in molecular structural parameters. For example, carboxylic acids show either the common dimer or the rare catemer patterns in their crystals. A study of the crystals of different substituted phenylpropionic acids reveals the formation of catemer is only possible if there is a neighbouring additional hydrogen bond located to the $\text{HOC}=\text{O}\cdots\text{HOC}=\text{O}$ unit.²⁰

Comparable dimer and catemer patterns were observed in the two polymorphs of the simple organic amide glycolanilide. The thermodynamically more stable polymorph presents a simple $\text{O}-\text{H}\cdots\text{O}=\text{C}$ hydrogen bonded chain structure (Fig. 4a). Interestingly, the amide $\text{N}-\text{H}$ bonds are not involved in hydrogen bonding in this polymorph. The metastable polymorph, in marked contrast, does incorporate $\text{N}-\text{H}$ as well as $\text{O}-\text{H}$ hydrogen binding interactions in its packing, resulting in a hydrogen bonded chain and an additional dimer pattern.²¹

High level modern synthesis and crystallisation techniques now make it possible to obtain new phases of polymorphic systems which help us to get a deeper insight into the formation of supramolecular synthons. The first polymorph (form I) of benzamide has been well known since 1959.²² The crystal packing is determined by robust $\text{C}=\text{O}\cdots\text{H}-\text{N}$ supramolecular synthons which form dimer units. In the last few years two more new polymorphs have been isolated by using *in situ* crystallization techniques (form II)²³ and an automated lab reactor equipped with online sensors that afford more controlled crystallisation conditions (form III).²⁴ The hydrogen bonding motif is very similar in all three solid phases. The significant differences in the observed packing motifs are the $\pi-\pi$ interaction of the benzyl moieties. Form I is dominated by shifted π stacks. In form II

and III herringbone patterns are observed. In form II pairs of hydrogen-bonded ribbons have a T-shaped arrangement, while in the new form III the herringbone pattern consists of single ribbons.

Effect of high pressure on supramolecular synthons

A survey of carboxylic acid structures showed that more than 95% of the mono-functional carboxylic acids form cyclic dimers and less than 5% form catemers.^{25–27} Katrusiak and co-workers undertook pressure-controlled aggregation of carboxylic acids²⁸ and showed that in the series of carboxylic acids $\text{R}-\text{COOH}$, where size of R is in the region of the changing preference for the dimer and

catemer hydrogen-bonded motifs formation (dimers for smaller R and catemers for larger ones), pressure can efficiently change this preference. For instance the low temperature structures of formic and acetic acids are isostructural and the carboxyl groups are linked into catemers by $[\text{syn}-\text{anti}, 0^\circ]$ hydrogen bonds, whereas at high pressure formic acid crystallises in a space group with two symmetry-independent molecules linked into catemers by $[\text{syn}-\text{syn}, 180^\circ]$ and $[\text{anti}-\text{anti}, 180^\circ]$ hydrogen bonds (Fig. 5).

Again, in the case of CBrClFCOOH , the catemer arrangement switches to the dimer form at 1.37 GPa. Different polymorphs of 2-chlorophenol and 4-fluorophenol were also obtained at high pressure (0.12 and 0.28 GPa).²⁹ The

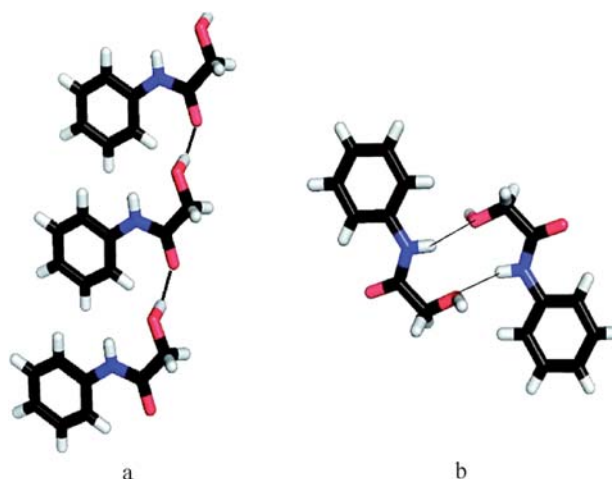


Fig. 4 Glycolanilide in (a) thermodynamically more stable catemer and (b) amide $\text{N}-\text{H}$ bonded dimer forms.

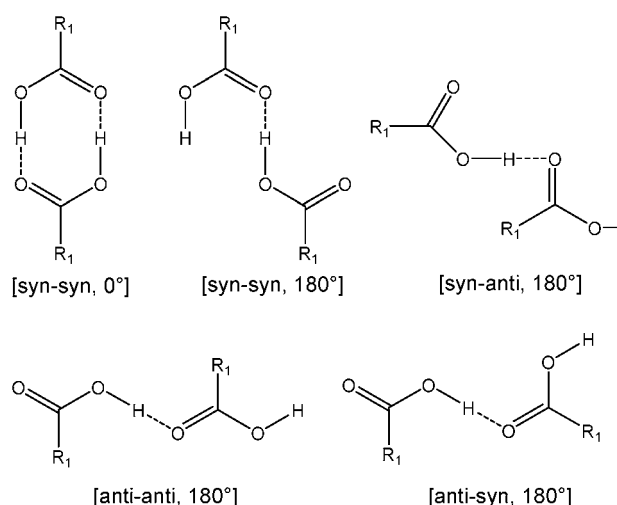


Fig. 5 Configurations of $\text{OH}\cdots\text{O}$ bonds linking carboxyl groups into planar aggregates.

structures of all phases are characterised by OH...OH hydrogen-bond formation. At ambient pressure six molecules of 4-fluorophenol interact *via* ...OH...OH... hydrogen bonds forming $R_6^6(12)$ ring motifs, which are connected together through F...H interactions (Fig. 6a). In contrast, at 0.28 GPa the molecules are connected by OH...OH hydrogen bonds into chains (Fig. 6b), which interact *via* F...F contacts to form layers. In the crystal of 2-chlorophenol the molecules interact *via* OH...OH hydrogen bonds to form helices, whereas the high pressure modification is determined by OH...OH chains. Crystallising in high-symmetry space groups, these two compounds conform to typical behaviour for bulky mono alcohols. Nonetheless, at high-pressure both compounds form zigzag chains arranged around 2_1 screw-axes, characteristic behaviour of small mono alcohols.

The recrystallisation of organic compounds from solution under high-

pressure conditions is shown to be a versatile method in the formation of new polymorphs. High pressure, as shown in the previous examples, is potentially a very valuable tool for tuning the balance between intermolecular interactions such as hydrogen bonding and van der Waals interactions. It often affects cell parameters such as axes lengths as well as intermolecular distances in comparison to the atmospheric pressure crystallisation.^{28–33} So the pressure of 1.1 GPa leads to the formation of a new polymorph of pyridine.³³

Are there characteristics of halogen bonding?

The nature of intermolecular interactions between halogen atoms, $X\cdots X$ ($X = \text{Cl}, \text{Br}, \text{and I}$), continues to be of topical interest due to their use for designing elements in crystal engineering.³⁴ In principle, halogen bonding has been known to chemists for nearly 150 years

and was briefly explored by chemists in the 1950s. But it is only in the last fifteen years that the scientific community has started to appreciate the importance of halogen bonding.³⁵ The geometrical characteristics of halogen-atom interactions ($\text{C}-\text{X}\cdots\text{X}-\text{C}$, $\text{X} = \text{Cl}, \text{Br}, \text{and I}$) have been classified into two categories (type-I and type-II) based on the geometrical $\text{C}-\text{X}\cdots\text{X}$ angles θ_1 and θ_2 (Fig. 7).^{36,37} Type-I interactions are symmetrical. These interactions are of the van der Waals type, such that the shortest among these are actually repulsive. Type-II interactions may be understood according to a model that assigns a positive polarisation in the polar region of the halogen atom and a negative polarisation in its equatorial region.^{36,38,39} But, a general assignment of well defined classes of halogen-substituted molecules to one of the two types is not always possible. For example the search in the Cambridge Structural Database for X-ray structures of halogen substituted benzenes with halogen...halogen interactions within the sum of the van der Waals radii (ref. 40) + 0.2 Å revealed ~100 structures.

The histogram (Fig. 8) shows that only a small part of these halogen...halogen contacts are shorter than the sum of the van der Waals radii. In an overview the interatomic angle can be topologically categorised by three regions in the histogram Fig. 9. Contacts in the region A with $\theta_2 < 90^\circ$ are ignored since other types of interactions are expected to play a dominant role in this region. About ~25 intermolecular contacts in the region C can be interpreted as type-I interactions with $\theta_1 \approx \theta_2$.

Nearly the same number of contacts, located in the region B with interatomic angle of θ_1 : 90° – 115° and θ_2 : 150° – 180° , could correspond to type-II. Considering ~25 contacts which are not located in one of the defined regions this study shows that in the case of halogen substituted benzenes the lengths and the angles of halogen-atom interactions are distributed randomly into the whole area. The existence and nature of the halogen bond have been thoroughly discussed in excellent reviews, highlights^{38,41,42} and concept papers by the Resnati–Metrangolo group. Recent theoretical studies using a high level of theory have confirmed the earlier studies on the structural and electronic

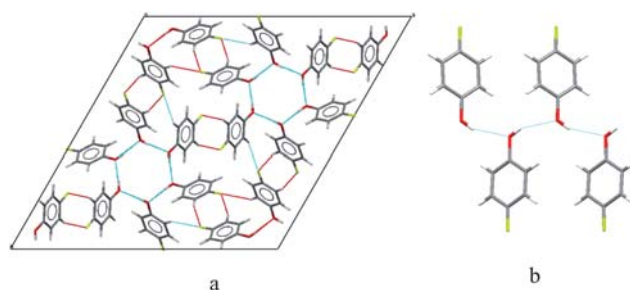


Fig. 6 Supramolecular arrangement in 4-fluorophenol (a) ...OH...OH... hydrogen bonded $R_6^6(12)$ ring motifs at 150 K and (b) OH...OH chains at 0.28 GPa.

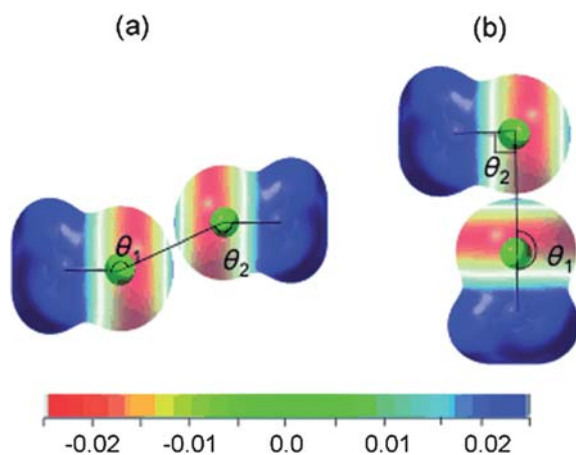


Fig. 7 Geometries of the two energy minima of chloromethane: (a) type-I $\theta_1 = \theta_2 \approx 150^\circ$, torsion angle ($\text{C}-\text{X}\cdots\text{X}-\text{C}$), $\phi = 180^\circ$; (b) type-II $\theta_2 = 180^\circ$ and $\theta_1 = 90^\circ$. The colour variation indicates the value of the calculated electrostatic potential.

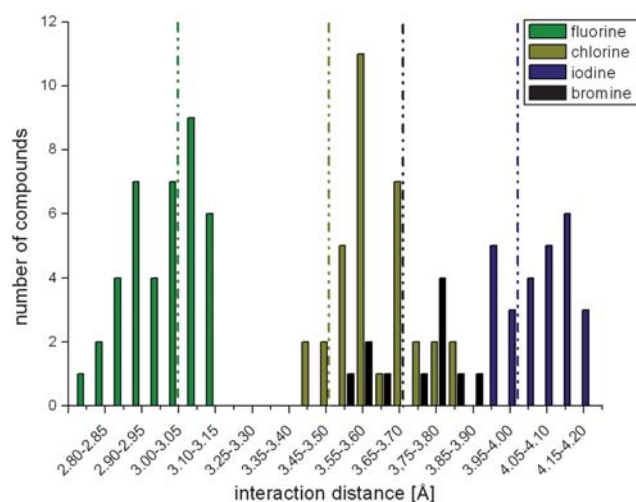


Fig. 8 Overview of CSD search for X-ray structures of halogen substituted benzenes with halogen...halogen interactions within the sum of the van der Waals radii (ref. 40) (---) + 0.2 Å (CSD-search: version 5.31, update Feb. 2010).

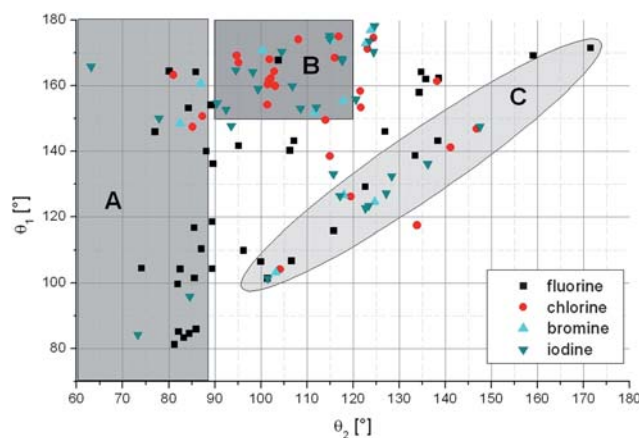


Fig. 9 Overview of the interatomic angle distribution θ_1/θ_2 , topologically categorised by three regions for ~100 halogen substituted benzenes (CSD-search: version 5.31, update Feb. 2010).

similarities between halogen and hydrogen bonding, yet have also indicated subtle differences reflected in some thermodynamic^{43,44} and spectroscopic properties.⁴⁵

The current importance of these topics is indicated by the choice of theme of the symposium held during an ACS Meeting in 2009: “Halogen Bonding: a World Parallel to Hydrogen Bonding”. The aim of the symposium was to draw the attention of the global scientific community to the huge and largely unexplored potential of halogen bonding by presenting its basic concepts and identifying and defining the boundaries of the field with a selection of the most interesting applications.³⁵ The significance of halogen bonding in biological systems has recently been attracting

attention thus making the concept of halogen bonding more widespread.^{45–47} Increasing experimental evidence indicates that organic chlorine and bromine atoms are favourably involved in discrete noncovalent dipolar protein–ligand interactions and thus contribute significantly to binding affinity in molecular recognition.^{38,46} Matter *et al.* present a systematic study of the hydrophobic aromatic protein where the Cl/Br... π -interaction was found to be the key factor for the identification of the potent and orally bioavailable molecule AVE-3247. These interactions have a characteristic geometry: the chlorine atom tends to reside on top of the plane of the neighbouring aromatic ring with the Cl–C bond approaching its aromatic plane from an angle of 60–90°.

Remarkably, this motif stabilises small-molecule crystal structures and protein–ligand complexes from multiple families, which indicates that this interaction is of a more general nature and not defined solely by the nature of the pocket or the protein in which the interaction takes place.⁴⁸

Ho and co-workers have investigated Holliday junctions (four-stranded DNA junctions) that are believed to be key structural intermediates during homologous recombination of DNA. When thymine is replaced by 5-bromouracil in the base sequence, the stacked-X form of a Holliday junction is stabilised as a result of short Br...O contacts (Fig. 10). A bromine atom at the inside crossover strand indicates formation of the halogen-bonded isomer; a bromine atom at the outside strand is indicative of the hydrogen-bonded isomer. This assay proved halogen bonding to be stronger than the analogous hydrogen bonding interaction in the same environment.⁴⁹

A survey of protein and nucleic acid crystal structures showed halogen-bonds are geometrically perpendicular to energetically independent hydrogen bonds that share a common carbonyl acceptor. This orthogonal relationship is accommodated by the in-plane and out-plane electronegative potentials which are differently populated by hydrogen- and halogen-bonds. This concept of orthogonal molecular interactions can be exploited for the rational design of halogenated ligands (Fig. 11).⁵⁰

Aggregation of organic fluorine compounds in solid state

Substitution of hydrogen with fluorine atom drastically changes the physical as well as chemical properties of a compound and the fluorine-containing compounds found to have wide applications in different aspects.⁵¹ A number of reports in the literature conclude that organic fluorine hardly accepts hydrogen bonds and does not contribute to the crystal packing. It was claimed by Dunitz and Taylor that fluorine, being the most electronegative element, hardly even forms C–H...F–C hydrogen bonds in contrast to the less electronegative N and O.⁵² A number of studies have been carried out in recent years that show fluorine generating different types of

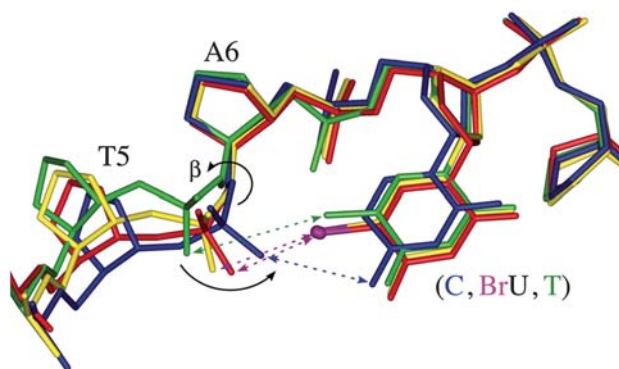


Fig. 10 The unusually short bond between bromine of 5-bromouracil and the phosphate oxygens in a four-stranded DNA junction is an example of non-covalent halogen bonding.

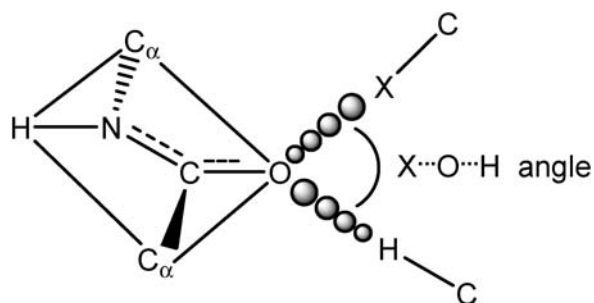


Fig. 11 Relative angle of approach of H- and X-bonds to a shared Lewis base (oxygen) for X = Cl, Br, and I.

packing motifs *via* C–H...F, C–F...F, and C–F...pi interactions, especially in the absence of strong hydrogen bond donors and/or acceptors such as N–H, O–H, C=O, *etc.*^{53–57}

C–H...F interactions are very weak and are controversially considered to belong to hydrogen bonds. Boese *et al.* defined four possible type of C–H...F synthons (Fig. 12) that are topologically similar to well known C–H...O and C–H...N hydrogen bonds.⁵² It has been shown on the range of investigated partially substituted fluorobenzenes that C–H...F interactions can be as important as C–H...N hydrogen bonds in stabilising specific crystal structures.

The comparison of 2,4,6-trifluoropyridine⁵⁸ with 1,3,5-trifluorobenzene and 1,3,5-triazine⁵³ proves this statement (Fig. 13). Interestingly, a dependency of the coplanar arrangement of pyridine molecules is observed with increasing number of fluorine substituents.⁵⁸ In contrast, a comparable planarisation effect depending on the number of fluorine substituents is not visible in fluorinated benzenes.

Hyla-Kryspin⁵⁹ showed that the electrostatic term in weak F...H hydrogen bridges can be even smaller than the dispersion term. Guru Row considers C–H...F–C contacts quantified *via* charge

density studies to belong to the hierarchy of hydrogen bonds.^{60,61}

He made the interesting observation of considerably stronger C_{sp}–H...F interaction, which overshadow the anticipated and very likely ≡C–H...O=C hydrogen bond in pentafluorophenyl prop-2-yl carbonate.

D'Oria and Novoa⁶² reinvestigated in 2008 the nature of the C–H...F interactions by CSD analysis. The strongest of the attractive C–H...F interactions was found to have strengths well beyond the –50 kcal mol^{–1} covalent limit (similar to what has also been found on O–H...O interactions). This was doubtful, however, and the authors themselves noted that in the total energy, the total van der Waals term dominated over the electrostatic term.

The question of whether F...F contacts are really interactions or whether they are caused by the close packing in the crystal is controversial. The nature of F...F interactions has still not been understood because fluorine is a very complex element from the supramolecular perspective.⁶³ It seems that fluorine would form C–H...F interactions rather than F...F contacts, whereas the heavier halogens seem to prefer the formation of halogen...halogen interactions.⁵³ According to Pauling's⁶⁴ principle fluorine has a low polarisability, so that the attractive interatomic dispersion forces are rather low and F...F interactions were rarely observed and if

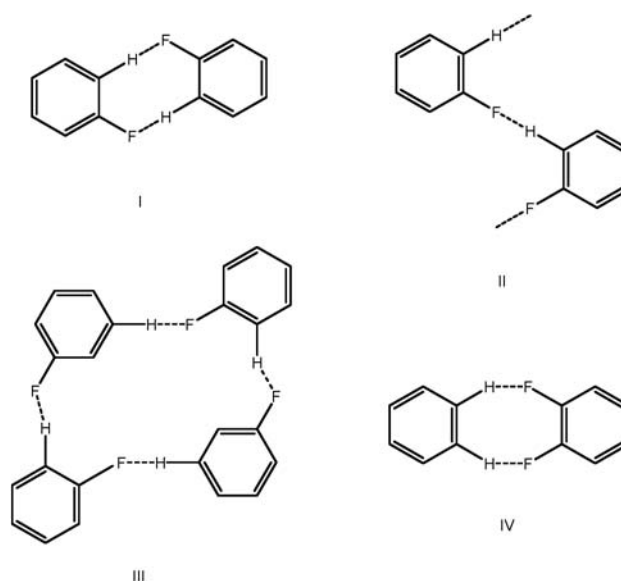


Fig. 12 Classification of supramolecular synthons based on C–F...H interactions.

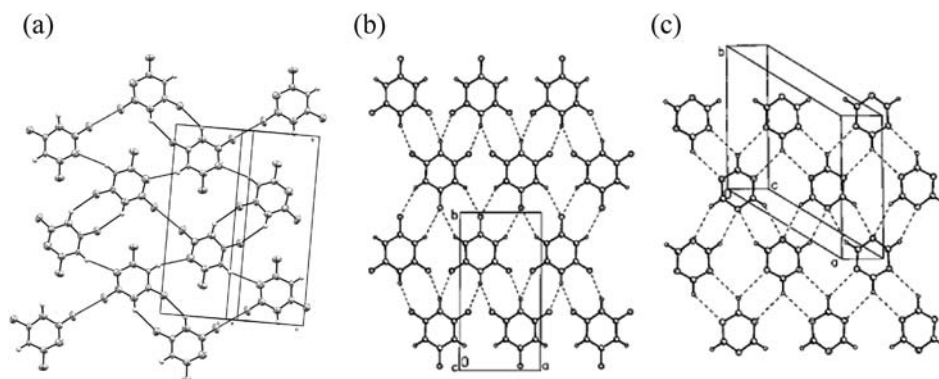


Fig. 13 Structural similarity between 2,4,6-trifluoropyridine (a), 1,3,5-trifluorobenzene (b) and 1,3,5-triazine (c).

present they were weak. CSD contains 788 compounds with $F\cdots F$ distances below $3.0\text{\AA}$, of which only 13 compounds do not have geometries with $\theta_1 \approx \theta_2$.⁵⁵

Hulliger determined crystal structures of some fluorinated benzophenones⁶⁵ where all possible interactions of the organic fluorine C–F including $F\cdots F$ and $F\cdots H$ are involved in the crystal structures depending on the number of fluorine atoms. In decafluoro-, 2,4,6,2',4',6'-hexafluoro- and 2,3,4,5,6-pentafluoro-benzophenone the atoms mainly involved in supramolecular interactions are those in the *ortho*- and *meta*-position of the aromatic rings though the *para*-positioned atoms were also found to be effective.

Schwarzer and Weber report for 18 fluorinated *N*-phenylmaleimides⁵⁵ and corresponding phthalimides an increase of $F\cdots F$ contacts with increasing the number of fluorine atoms in the molecule, while the $C-H\cdots O$ and $C-H\cdots F$ contacts decrease. Nevertheless these contacts are supposed to be rather secondary and mostly determined by the packing. However, the examples of $F\cdots F$ contacts in 2,3,6-trifluoropyridine,⁵⁸ pentafluoropyridine,³⁰ highly fluorinated benzophenones⁶⁵ and *N*-phenylmaleimides⁵⁵ may give an indication that a bottom limit of fluorine atoms seems to be necessary for successful competition with the hydrogen type of interactions. In contrast the groups of Guru Row⁶⁶ and Bagryanskaya⁵⁶ consider the existence of $F\cdots F$ nonvalent contacts likely, having the effect of stabilising the packing.

Supramolecular synthons with $\pi\cdots\pi$ or $C-F\cdots\pi$ interactions

Bagryanskaya *et al.*^{67,68} showed some years earlier that the basic supramolecular

motifs in the homomolecular crystals of partially fluorinated heteroarenes are formed by $\pi\cdots\pi$ and $C-F\cdots\pi$ stacking interactions. In most cases there are also short $F\cdots F$ contacts between the stacks. Besides, 1,2,3,4-tetrafluoronaphthalene depicted the very rare case of co-existing arene–arene, arene–polyfluoroarene and polyfluoroarene–polyfluoroarene $\pi\cdots\pi$ interactions within the infinite π -stacks. It was speculated that such a type of structure might be suitable, for example, for both hole- and electron-transporting activity.

The $\pi\cdots\pi$ stacking and edge–face contacts between aromatic groups occupy prominent positions among non-covalent interactions.⁶⁹ Fluorine substitution is known to modulate aromatic/aromatic interactions by affecting the HOMO–LUMO gap, which leads to a strong propensity for the molecules to aggregate.⁷⁰ Benzene–perfluorobenzene with its stacks is the most well known example of such an interaction where the crystal-packing motif reflects the maximisation of favourable quadrupole–quadrupole interactions and the minimisation of unfavourable electrostatic repulsions (for example, edge-to-face arrangements).⁶⁹ The interaction between phenyl and perfluorophenyl fulfills the conditions of structural predictability for consideration as a supramolecular synthon.³ In contrast, of 18 fluorinated compounds, a $\pi\cdots\pi$ (F) interaction was only found in two exceptional cases for dimer formation, contradicting the common opinion of $\pi\cdots\pi$ stacking as being a robust supramolecular synthon.⁵⁵

In general $\pi\cdots\pi$ interactions between aromatic moieties are useful in crystal engineering and supramolecular chemistry. They can be used to manipulate the

organisation of molecular components in the solid state. Grimme⁷¹ considers that the π orbitals do not function as in conventional overlap-driven covalent bonding. He engaged himself with the question of whether the term “ $\pi\cdots\pi$ -interaction” makes sense from a theoretical point of view.

Amazingly, the three polymorphs of benzamide²⁴ do not differ in the dominating hydrogen bonding motifs, but in the π – π interaction of the benzyl moieties. The favoured π – π interaction motifs are shifted π -stacks (form I) and T-shaped interactions resulting in herringbone-type motifs (both metastable forms II and III).

Champness *et al.* investigated a range of similar 1,2,4,5-tetrazine rings.⁷² In all the examples, with one exception, the molecules assembled into stacks. The observed interactions are between tetrazine rings and the end group rings (Fig. 14). Such an arrangement allows for neighbouring molecules within a stack to have two $\pi\cdots\pi$ interactions, between relatively electron-rich and electron-poor rings. The packing arrangement of the molecules is affected by an interplay of competing weak forces, most notably $\pi\cdots\pi$ interactions and $C-H\cdots N$ hydrogen bonding. Only in the case of 3,6-diphenyl-1,2,4,5-tetrazine, no π – π interactions are observed. These results illustrate that π – π interactions will not always be adopted by aromatic molecules, particularly when the molecule includes equally or almost equally electron-deficient or indeed electron-rich rings.

Anion– π interactions, having $C-H\cdots$ anion hydrogen bonding as a predominant interaction motif with charge-neutral π systems, are considered to be extremely rare in crystal structures.

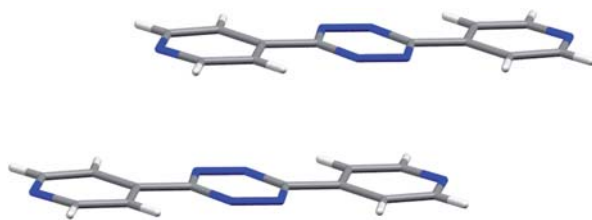


Fig. 14 The crystal structure of 4,4'-dptz shows interactions between tetrazine rings and the end group rings, undergoing two $\pi \cdots \pi$ interactions.

Reedijk *et al.*⁷³ demonstrated, by a thorough look at the crystal structures deposited in the CSD, that the anion- π contacts in fact are quite common supramolecular bonding interactions in solid-state structures. A number of common anions have been found to interact strongly with (*N*-hetero) aromatic six-membered rings.

Hay and Custelcean⁷⁴ looked for contacts between arenes and the contacting species and could not even find one clear example of an anion- π interaction with a charge-neutral arene in the entire CSD database. Thus, they conclude that intrinsic examples of anion- π interactions with uncharged arenes are extraordinary.

Chloro-, bromo- and iodo-substituted 5,5-bisdiazodipyrromethanes⁷⁵ form interlocked dimers through quadruple N-H $\cdots$ N hydrogen bonds, but the halogen $\cdots\pi$ synthons also provide a significant contribution to the resulting supramolecular motifs. In fact, Nangia and co-workers⁷⁶ have found that the halogen/ π interactions play a key role in stabilising the host-guest complexes. Bishop and co-workers⁷⁷ have witnessed the important role of Cl/ π and Br/ π interactions on guest inclusion of the diquinoline host. Vysotsky and co-workers⁷⁸ have also found that halogen/ π interactions are critical for directing the assemblies of tri-bromo and tri-iodo trityl alcohol hosts.

X $\cdots$ Y synthons with X and Y = halogen, nitro, ethynyl or cyano

The hydrogen atom is the most common electron acceptor site and hydrogen bonding is by far the most frequently occurring noncovalent interaction in recognition processes. But a number of weaker and softer interactions, such as X $\cdots$ Y synthons with X and Y = halogen, nitro, ethynyl or cyano, have also been

shown to play a role in structure stabilisation.⁷⁹ Several studies on halogen-substituted nitrobenzenes and benzonitriles show that the different possibilities of recognition motifs of X $\cdots$ X and X $\cdots$ Y interactions depend on the substitution pattern on the benzene ring.^{80–82} This is to be observed in particular with the weaker synthons on participation of iodine. In the crystal packing of *ortho*- and *meta*-iodosubstituted benzonitriles weak and soft supramolecular synthons of I $\cdots$ I zigzag chains dominate. On the other hand *para*-iodobenzonitrile, which exhibits a lower dipole moment, shows the formation of linear CN $\cdots$ I arrays (Fig. 15).

The question is whether weakly directing substituents influence the aggregation of the aromatic molecules in the solid. For example the mono-substituted benzenes, fluorobenzene⁵³ and benzonitrile,⁸³ crystallise isostructural and X-ray diffraction analysis of *o*-, *m*-, and *p*-nitrotoluenes shows that these compounds have similar structural parameters close to those found in toluene and nitrobenzene. In crystals of the isomers studied, molecules are linked

in sheets by short N-O $\cdots$ H interactions (Fig. 16). All studied compounds have been examined with respect to the propensity of generating molecular packing motifs particularly involving N-O $\cdots$ H and π - π staking interactions. In fact, no similarities between packing modes of these compounds were found.⁸⁴

But supramolecular synthons based on weak intermolecular interactions like C-H $\cdots$ N hydrogen bonds are robust and they can be used in the creation of structures with different topologies. The crystal structures of three ethynylpyridines and an ethynylpyrimidine show that, depending on the substitution pattern of the aromatic heterocycles, short, linear hydrogen bonds formed by ethynyl groups can be used to generate networks with predictable topologies (Fig. 17).⁸⁵ Remarkable self-associations based on weak intermolecular interactions have been reported for several crystal structures containing orthogonal dipolar C=O $\cdots$ C=O interactions.^{86,87} Alloxan is a prominent example of such a compound with an unusually high melting point, high crystal density and also the absence of conventional hydrogen bonds in the crystal structure.⁸⁸ The crystal structure of alloxan has fascinated scientists since the X-ray study was published in 1964.⁶³ In contrast to earlier studies, a recent neutron powder diffraction investigation⁸⁹ has shown that the crystal structure of deuterated alloxan transforms to a new orthorhombic isotopomorphic phase

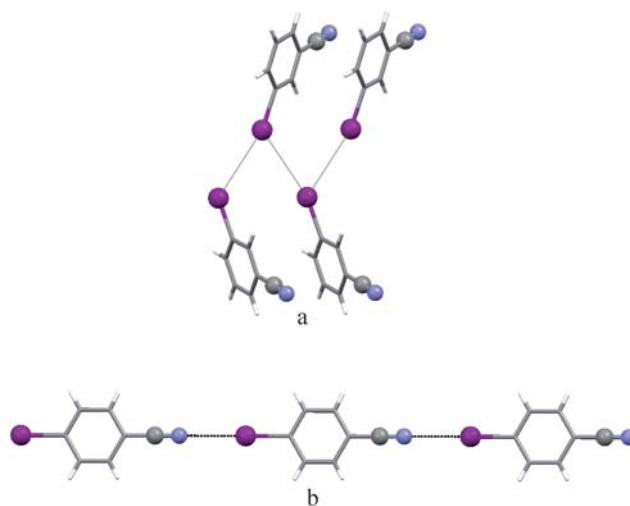


Fig. 15 The crystal structure of *meta*-iodobenzonitrile is determined by I $\cdots$ I zigzag chains (a) opposite to CN $\cdots$ I bonded strands of *para*-iodobenzonitrile (b).

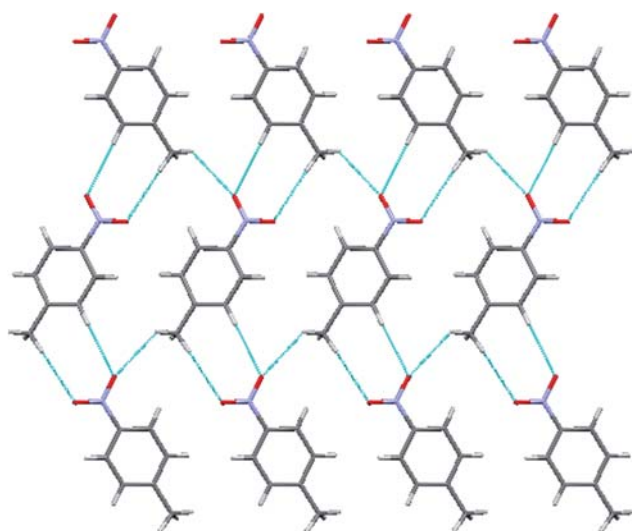


Fig. 16 In *para*-nitrotoluene the molecules are linked in sheets by short N–O...H interactions.

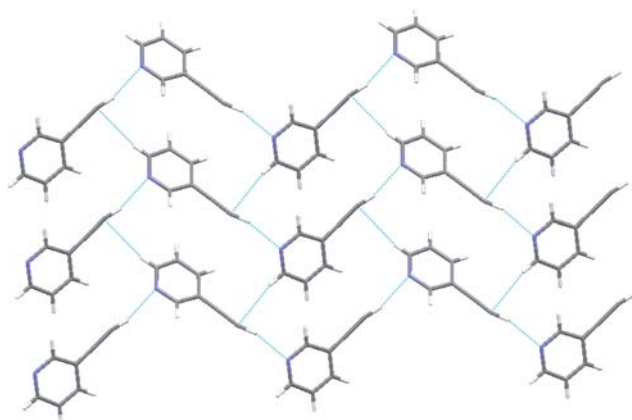


Fig. 17 The crystal structure of 3-ethynylpyridine is shown as a networks of supramolecular synthons based on weak inter-molecular C–H...N hydrogen bonds.

below 35 K. The two phases are almost superimposable.

There are two main types of inter-molecular interactions: quite similar CO...CO contacts in both phases but different NH...O contacts. Conspicuous is the comparable phase transition of pyridine and pyridine-*N*-oxide depending on the H/D-exchange investigated by Boese *et al.*³³ and our group.⁹⁰ Therefore, weak crystal structure determining intermolecular interactions can be easily affected by simple H/D exchange and consequently new isotopomorphic phases can be achieved.

Crystal structure prediction (CSP)

One important useful consequence of the supramolecular concept should be the desired predictability of crystal structures. The importance of crystal structure

prediction (CSP) has been recognized through the “blind tests” conducted by the Cambridge Crystallographic Data Centre (CCDC).^{91,92} The results of the fourth blind test in 2007 (CSP2007) demonstrate that at least for the small and fairly rigid types of molecules successful prediction is actually possible within the restrictions of the blind test.

Acrolein was chosen as the blind test target for small rigid molecules (Fig. 18a). Acrolein crystallises in the orthorhombic space group *Pbca* with *Z'* = 1. The crystal structure is determined by weak interactions and each molecule is found to make six short C–H...O contacts with neighbouring molecules.

An additional small molecule, the halogenated arene, 1,3-dibromo-2-chloro-5-fluorobenzene (Fig. 18b), was selected as molecule containing functional groups that present a challenge for modelling methods. There are two Br...Br close contact distances of 3.55 and 3.63 Å in the observed structure. F atoms form nearly head-to-head close contacts between coplanar molecules with a F...F distance of 2.87 Å. Most of the fourteen participating research groups attempted predictions for these molecules. Four of them predicted the observed structures within their three predictions with RMSD16 deviations from the experimentally determined structure of less than 0.2–0.4 Å and root-mean-squared errors in the unit-cell lengths of less than 2.4–3% for acrolein and 1,3-dibromo-2-chloro-5-fluorobenzene.

The prediction of flexible crystal structures is more difficult. For example 6-amino-2-phenylsulfonylimino-11,2-dihydropyridine was part of the second blind test in 2001 (Fig. 19). None of the participants were able to predict the crystal structure (form I) which contains the two-point synthon B. Further study yielded two more polymorphs. Form II shows a dimer hydrogen-bonding four-point synthon A as building block.^{93,94} Form III exhibits a catemer two-point synthon B, similar to form I with N–H...N and N–H...O hydrogen bonds.⁹⁵ By solvent-assisted polymorphic transformation experiments, Matzger and Roy⁹⁵ have shown form II and form III can be converted to the thermodynamically favored form I.

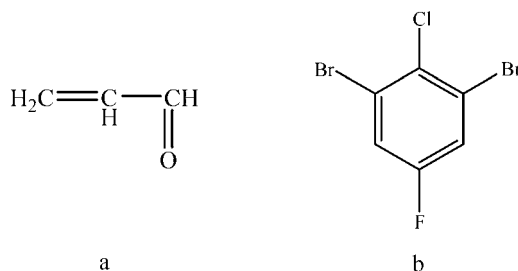


Fig. 18 Molecular structure of acrolein (a) and 1,3-dibromo-2-chloro-5-fluorobenzene (b).

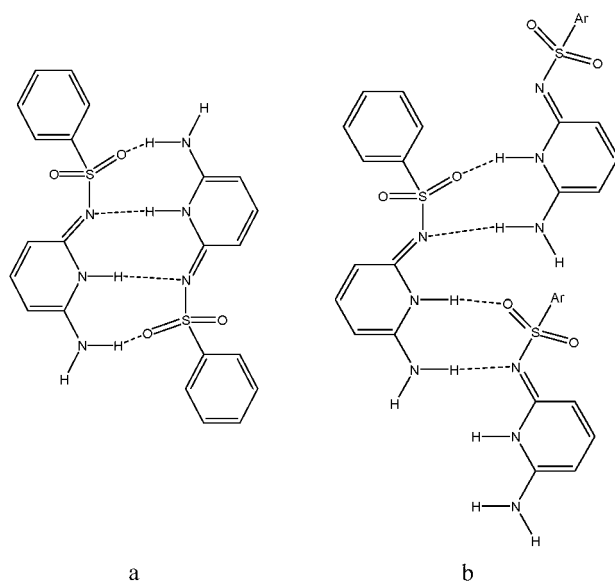


Fig. 19 The four-point synthon A (a) and two-point synthon B (b) of 6-amino-2-phenylsulfonylimino-11,2-dihydropyridine, used for the second blind test in 2001.

In contrast, the limits of CSP have been shown in the experimental and theoretical study of isomers of dichloronitrobenzene and 3,4-dinitrochlorobenzene.⁹⁶ Improvements in the computational model are necessary to determine the most thermodynamically favourable structure of these molecules. The modelling of the nitro group flexibility is essential: the phenyl-nitro dihedral angle can be readily deformed by packing forces, varying by around as much as 30° for the same adjacent substituent. Taking account of the nitro group, rotation in crystal structure prediction has been deemed “a very problematic task”⁹⁷ as it appears to have a significant influence on the packing mode of nitrobenzene derivatives.

The experimental study of dichloronitrobenzene shows the presence of

C–Cl⋯O(N) interactions. In contrast to iodonitrobenzenes, which tend to form bifurcated C–I⋯O₂(N) interactions,⁹⁸ these chloronitrobenzenes only form monocoordinate C–Cl⋯O(N) interactions with the C–Cl⋯O angle approaching linearity (145–172°) in accordance with a general trend. Such C–Cl⋯O short contacts are clearly formed in preference to Cl⋯Cl interactions which have been widely observed and analysed⁹⁹ in the chlorobenzenes.^{100,101} The study shows that it is not currently possible to correlate the crystallisation outcome with the functional groups present in any general way.

When the kinetic form is the one obtained experimentally, Desiraju *et al.* have suggested a knowledge-based alternative, the supramolecular synthon approach to

CSP^{92,102} using “small” and “large” synthons. This synthon based CSP works well for most rigid isomeric methylaminophenols and simple unsubstituted aminophenols.^{103,104} Fig. 20 shows synthons in the crystal structures of these compounds. Synthon I is classified as small and IV may be considered to be large. Synthons II and III are intermediate in complexity. The absence of a small synthon in a predicted structure is a negative factor and justifies its down-ranking or elimination. The presence of a large synthon in a predicted structure is a positive factor and is grounds for its up-ranking. The comparison of experimental and predicted structures shows that a Me/Et substitutional change has a serious effect on the crystal packing whereas a phenyl/naphthyl substitutional change does not.¹⁰³

While crystal structure predictions almost always generate more hypothetical structures than known polymorphs, it seems likely that some of the previously unobserved structures from CSP studies should be accessible through changes in the crystallisation conditions like the high-pressure crystallisation technique. For example the predicted polymorphs at low- and high-pressure of 2-chlorophenol and 4-fluorophenol were found to have an energy difference of only a few kJ mol^{−1}.²⁹

Weak intermolecular interactions—the borders of the concept of supramolecular synthons

In this context it is necessary to talk about chemical bonding. One theoretical approach based on *ab initio* schemes is derived from wave function and electron

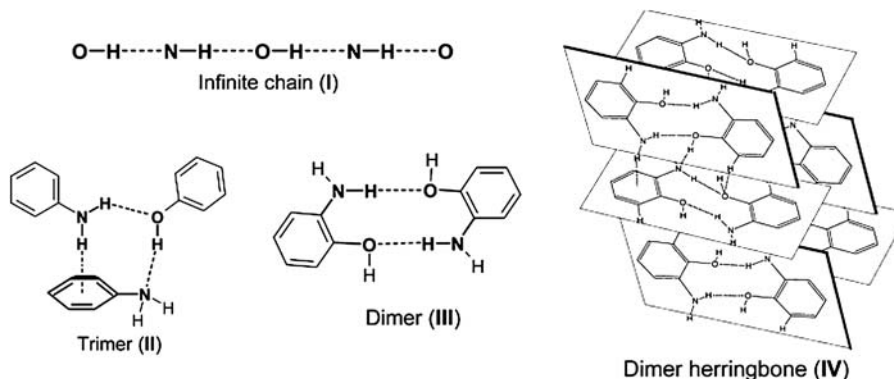


Fig. 20 “Small” and “large” supramolecular synthons of aminophenols.

density calculations. But the problem is the lack of a quantum mechanical “bond operator” that would provide the desired answer as a conventional expectation value. One basic assumption should be that the existence of a chemical bond must be related to some observable; in other words, chemical bonding must have an effect on measurable properties of the system. Any definition of bonding that has no real consequences is meaningless and must be abandoned.¹⁰⁴ Bader’s “Theory of Atoms in Molecules” (AIM)¹⁰⁵ is certainly one of the most fundamental bond concepts. Herein bonds are defined by a bond path (BP) and a bond critical point (BCP). The BP is a line in space linking the nuclei of bonded atoms where the electron density is a maximum with respect to any neighbouring line. It is assumed that a BP is necessary for chemical bonding regardless of its nature.¹⁰⁶ The BCP represents a stationary point in the electron density and is a minimum along the BP but a maximum in directions perpendicular to the bond path. The values of the density, its Laplacian, or energy-based properties at the BCP can be used to characterise the nature of the interaction in some cases.¹⁰⁵

Attention has begun to focus on the link between structural synthons and growth synthons present in solution as a consequence of thermodynamically driven self-association equilibria. Using proton NMR technique, Spitaleri *et al.* have noted a direct correlation between dimers in the solution phase and those in the resulting crystal in the case of sulfamerazine. The process of molecular self-assembly has been studied in concentrated solutions of carboxylic acids using FTIR spectroscopy.¹⁰⁷ The acids have been chosen on the basis of their differing crystal chemistries, as reflected in observed hydrogen-bonding motifs. In the cases of benzoic acid and tetrolic acid, a link between the growth synthon and the structural synthon is apparent. It appears that benzoic acid dimers are present in polar and nonpolar solution. The observed growth synthon of tetrolic acid depends on the solvent. At 20 °C, chloroform solutions which are rich in dimers yield crystals of the dimer-based α polymorph directly, while ethanol solutions, rich in nondimeric aggregates, yield the β form. Additive-induced

polymorphism has always been of interest to solid-state chemists for fundamental and practical reasons.¹⁰⁸

Desiraju and Boese *et al.*¹⁰⁹ obtained a noncentrosymmetric polymorph of 1,3,5-trinitrobenzene (TNB) by introducing the additive trisindane (TI), which is structurally similar to the molecule of interest. TI interacts with TNB, but not too well. It seems to be the critical factor that determines the appearance of new polymorphs. The major distinction of the two polymorphs is the centrosymmetric and polar arrangements of the molecule. Both polymorphs crystallize with $Z' = 2$.

Another important and amazing example is pyridine. This compound is one of the simplest heteroaromatics, although its crystal structure, having four¹¹⁰ independent molecules in the asymmetric unit, is unusually complicated ($Z' = 4$). Isotopic substitution is the smallest possible alteration of a molecule. Nevertheless this can support formation of polymorphs for systems which are otherwise monomorphic. For pyridine, an isotopic substitution leads to the formation of a new polymorph. Recrystallisation from pentane allows the formation of a new stable polymorph of penta-deuteropyridine (Fig. 21).³³ Besides, recent investigations of Podsiadlo *et al.* indicate that the variety of pyridine forms appears to be driven thermodynamically.¹¹¹

The most prominent interactions in both phases are $\text{CH}\cdots\text{N}$ and $\text{C-H}\cdots\pi$ contacts. In general H/D exchange is seen as a non-dominating parameter for the structural or physical properties of a compound. Our own recent investigations of partially deuterio-substituted

pyridine-*N*-oxides⁹⁰ yield two different polymorphs for a threefold deuterated compound. Both the low-temperature ($Z' = 2$) and the high-temperature ($Z' = 0.5$) phases are stable and reversible (Fig. 22).

Such examples of isotopic polymorphism under varying conditions of temperature and pressure will form particularly sensitive test cases when dynamics are introduced into crystal structure prediction methodologies. Desiraju defines structure with $Z' > 1$ as crystals “on the way” in the sense of being frozen in some high energy kinetic forms.¹¹² He suggests further that the probability of appearance of a high Z' structure will depend largely on the crystallisation temperature. In those cases where kinetic and thermodynamic forms are one and the same, $Z' > 1$ is not possible.

Brock and Duncan have extensively surveyed a great range of alcohols and phenols.¹¹³ According to their explanation, the tendency of these hydroxy compounds (ROH) to form cooperative hydrogen bonded chains of the $\text{O-H}\cdots\text{O-H}$ type is countered by the steric demands of the R-groups, and high Z' is one of the outcomes. Nangia mentioned that the observations of hydrogen bond strength/high Z' correlation in $\text{O-H}\cdots\text{O}$ bonds in multiple crystallographically distinct molecules of the same chemical entry ($Z' > 1$) point toward the dominance of enthalpy in such crystallisations.¹¹⁴

Desiraju interpreted the $\text{O-H}\cdots\text{O-H}\cdots\text{O-H}$ trimer synthon in the high Z' polymorph of pentafluorophenol^{115,116} as an “incomplete” version of the $\text{OH}\cdots\text{OH}\cdots\text{OH}$ chains in the low Z' polymorph

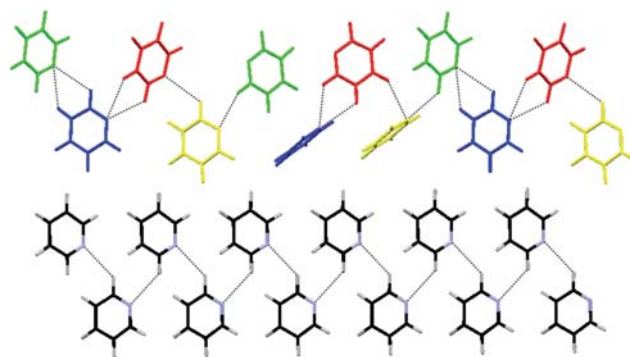


Fig. 21 The relationship between the crystal structures of pyridine (top) and deuteropyridine (bottom). The $\text{CH}\cdots\pi$ interaction in the middle of the phase-I chain (top) is converted into a $\text{CH}\cdots\text{N}$ contact in phase-II.

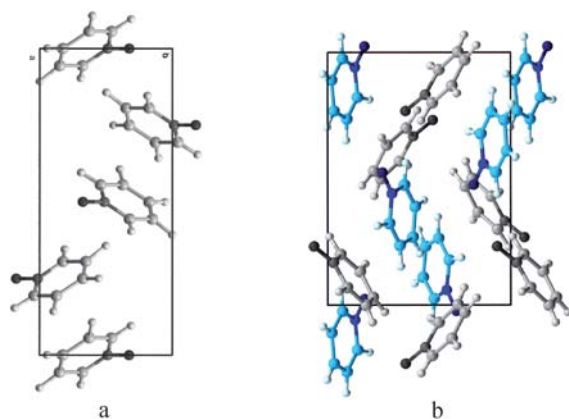


Fig. 22 Crystal structure of the partial deuterated 2,3,6-trideuteropyridine-*N*-oxide: (a) $-30\text{ }^{\circ}\text{C}$ in the space group $P4_12_12$ (phase I); (b) $-50\text{ }^{\circ}\text{C}$ in the space group $P2_12_12$ (phase II).

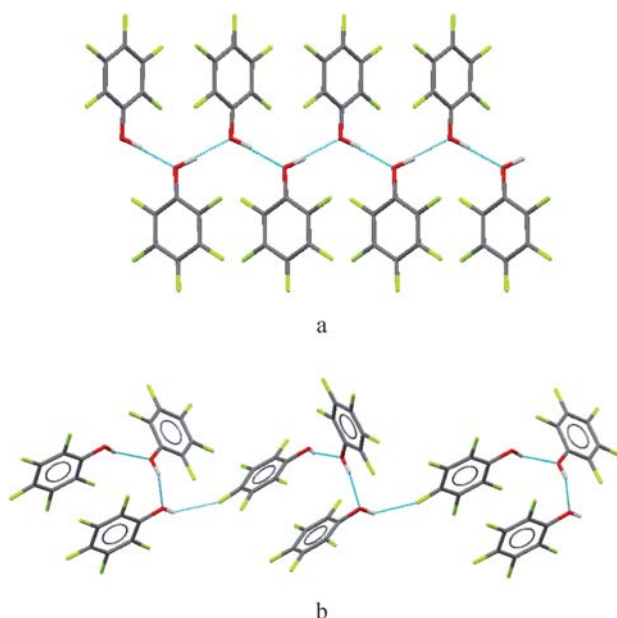


Fig. 23 Crystal structure of pentafluorophenols: (a) form I with infinite $\cdots\text{OH}\cdots\text{OH}\cdots\text{OH}\cdots$ chains ($Z' = 1$) and (b) form II with discrete $\text{OH}\cdots\text{OH}\cdots\text{OH}$ trimers ($Z' = 3$).

(Fig. 23). Pentafluorophenol ($Z' = 3$) and *trans*-1,4-bis(phenylethynyl)cyclohexane-1,4-diol ($Z' = 8$) have respectively two polymorphs, one of which has a high value of Z' .¹¹⁵ Different crystallisation behaviour was found for both compounds depending on the crystallisation conditions.

Conclusions

In this article we have outlined some current developments in the field of supramolecular synthons. In recent years the concept of supramolecular synthons has been receiving ever increasing attention in various fields. This trend is manifested in the exploding number of articles

published, not only in the specialist journals, *Crystal Growth and Design* and *CrystEngComm*. The fact that robust supramolecular synthons can be used successfully for crystal engineering purposes has especially received attention. The borders of the concept become visible if the smallest structural molecule alteration (e.g. H/D and H/F exchange) or different crystallisation techniques can influence intermolecular interactions and the aggregation of molecules in the solid state. In order to understand and predict crystal structures where weak intermolecular interactions could have a stabilising role, both a greater number of crystal structure investigations with highly developed crystallisation techniques and

also improvements in the computational models will be necessary.

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